MATH 4360 HOMEWORK ASSIGNMENT 5

DUE ON THURSDAY 28 SEPTEMBER 2023

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

(1) Consider $z = 5 + 5i$ and $w = 1 - 3i$ in the ring $\mathbb{Z}[i]$.
   (a) Factor w as a product of primes.
   (b) Find a greatest common divisor of z and w.

(2) Factor $x^4 - 8x^2 + 15$ as a product of irreducible polynomials over $\mathbb{Q}$.

(3) Show that $\mathbb{Z}[i\sqrt{3}]$ is not a Euclidian domain.

(4) Let R be a Euclidian domain and a and u be non-zero elements in R. Show that $v(au) = v(a)$ holds if and only if u is a unit in R. [*Hint: R is PID.*]